

Performance Analysis of Machine Learning Models for Predicting Return Loss in 5G Microstrip Patch Array Antenna Design

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Abstract—This work explores the integration of machine learning models to predict the return loss of a microstrip antenna operating at 3.5 GHz in the sub-6 GHz 5G frequency band. 11 (eleven) design parameters are varied to create a dataset of 380 samples. Eight regression models are used including Random Forest Regression, Decision Tree Regression, linear regression, and Extreme Gradient Boosting. Hyperparameter tuning is conducted via grid search and randomized search with cross-validation. Seven performance metrics, Mean square error, Mean absolute error, R-square, and Variance score, are used to evaluate the model. Random forest regression provides the best result with an MAE of 9.59%, R-square of 90.59%, and variance of 93.49%. This study also utilizes residual and feature importance analysis to understand the model performance better.

Keywords—Machine Learning, Prediction, Return Loss, Microstrip Patch Antenna, Hyperparameter Tuning.

I. INTRODUCTION

Machine Learning (ML) is critical to mimicking different complex tasks in a more straightforward method in the Artificial Intelligence (AI) era. Designing antennas can be automated by utilizing machine learning techniques. Antenna design requires manual parameter creation and tuning parameters using any electromagnetic modeling solution to get the required performance. However, this manual parameter tweaking or performance study of the designed antenna on a different parameter than the planned one will be less time-consuming and laborious if it can be trained in any ML model to its merit [1].

The antenna design domain can employ machine learning algorithms in two ways: performance prediction and antenna design parameter optimization. Work completed on performance prediction is focused here since machine learning approaches have been used to forecast the performance of a 5G band microstrip patch antenna. Various research has been done on the performance prediction of the antenna using ML involving resonance frequency, gain, bandwidth, and return loss prediction. The frequency and directivity of a quasi-Yagi antenna for UMTS 2100 MHz LTE were predicted using seven techniques, including ridge regression and linear regression, with ridge regression exhibiting superior accuracy in directivity prediction and linear regression in frequency accuracy [2]. In addition, a quasi-Yagi antenna's gain and directivity were

predicted using eight models, including ridge regression and Gaussian process regression, with a 98% accuracy rate in gain predictions [3]. To forecast bandwidth and resonant frequency for Yagi-Uda antennas, other research used artificial neural networks, decision tree regression, random forest regression, and XGB [4]. Random forest regression demonstrated over 95% accuracy in gain prediction in one investigation [5], while another involved six ML models to predict the gain of a 5G 28 GHz MIMO antenna [6]. To predict the S11 parameter of four antennas, four machine learning models—decision tree, random forest, XGB, and KNN—were used. Performance was assessed using MAE, MSE, and R^2 and comparing the predicted with actual S11 data points for each antenna [7]. The return loss of an eye bolt-shaped antenna was predicted using random forest, XGB, decision tree regression, and linear regression models; XGB performed better than the other models based on variance score, MAE, RMSE, and MSE, but no prediction visualization was given [8]. A further study used KNN, XGBoost, decision tree, and random forest models to forecast the return loss of THz antennas meant for 6G communication; random forest achieved the most excellent accuracy of 82% and the lowest MSE of 3.816 [9]. Furthermore, XGB regression was better than random forest, decision tree, and linear regression in forecasting return loss for the eye bolt-shaped antenna when assessed using variance score, MAE, RMSE, and MSE [1]. To predict return loss and compare the outcomes across various ML models, a tri-band antenna was eventually designed, and datasets were generated to train models for linear regression, decision tree regression, support vector machines, Gaussian process regression, efficient linear regression, and kernel regression [10].

While previous studies have employed various machine learning models to predict antenna performance metrics like gain, bandwidth, resonant frequency, and return loss, they often lacked comprehensive hyperparameter tuning and thorough performance analysis methodologies, such as feature importance and residual analysis; this research aims to fill these gaps by utilizing eight machine learning models with advanced hyperparameter optimization techniques and a broader array of evaluation metrics to predict accurately the return loss of a microstrip patch antenna operating at 3.5 GHz in the 5G

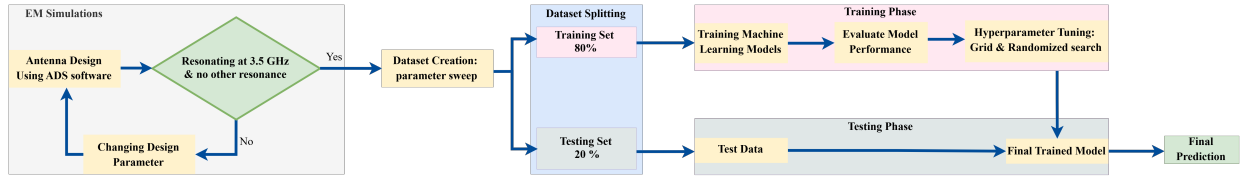


Fig. 1: Block diagram of the proposed methodology showing initial antenna design and machine learning technique.

frequency band.

The main contribution is summarized into three points:

- **Diverse ML Model Implementation:** To improve the accuracy of return loss predictions for a microstrip patch antenna at 3.5 GHz in the 5G band, this study uses eight machine learning models.
- **Enhanced Hyperparameter Optimization:** To achieve optimal model performance, this study uses sophisticated hyperparameter tuning methods, such as randomized and greedy searches combined with cross-validation.
- **Comprehensive Performance Assessment:** To learn more about the effectiveness and precision of the model, a variety of evaluation measures are employed in conjunction with feature importance and residual studies.

II. METHODOLOGY

The primary objective of this work is to use machine learning to forecast the return loss of the microstrip patch antenna that has been designed. The work is divided into two main parts: antenna design and prediction of return loss implementing machine learning techniques. An illustration of the entire process is provided in Fig. 1.

A. Antenna Design

A 2×2 microstrip patch antenna was designed [11] using Teflon glass fiber substrate separating ground and patch. Two 50Ω microstrip feed lines feed the entire patch network. The patches are connected via two quarter-wave transformers. A magic-T formation with a slot in the ground connects these transformers to the feed line. The dimension and parameter of the antenna are given in Table I and Fig 2.

This initial design method involves a trial-and-error approach to finding an optimum level by checking two things: it resonates at 3.5 GHz and has no other resonance. This antenna shows good impedance matching with a return loss of -43 dB and a gain of 13 dBi at 3.5 GHz. The return loss of the designed antenna is shown in Fig 3.

B. Dataset Creation

A dataset is created by changing the design parameter of the initial best-designed antenna up and below their best parameter value. The variation range, along with the variation steps, is presented in Table II. The length of the transmission line varied from $\lambda/4$ to $3\lambda/4$ approximately, and the stub length varied near $\lambda/4$. The other parameters remain at their initial best value when one parameter changes. By this process, a total of 380 data samples are created.

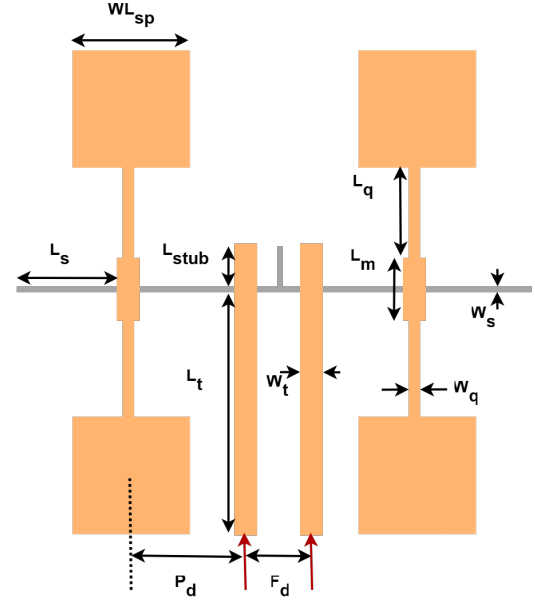


Fig. 2: Designed antenna with EM Simulation

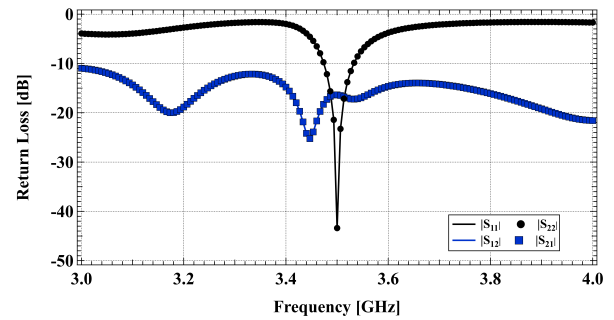


Fig. 3: Return Loss of the designed antenna.

Each sample has 11 design parameters, $X \in \mathbb{R}^{380 \times 11}$, and 101 return loss values, $Y \in \mathbb{R}^{380 \times 101}$. Various machine learning algorithms predict the return loss based on the design parameters. To do this, the dataset is divided into training (80%) and testing (20%). Overall training and test data are displayed in Fig 4.

C. Machine Learning Model Implementation

Let $X = [x_1, x_2, \dots, x_{11}]$ represent the design parameters, and $Y = [y_1, y_2, \dots, y_{101}]$ represent the corresponding return

TABLE I: Antenna design parameters

Parameter	Value	Parameter	Value
Single patch dimension, WL_{sp}	$30 \times 30 \text{ mm}^2$	Substrate thickness	0.8 mm
Microstrip line width, $W_t=W_m$	2.6 mm	Patch thickness	0.035 mm
Slot-line width, W_s	0.2 mm	Relative dielectric constant, ϵ_r	2.15
Substrate material	Teflon glass fiber	Quarter wave transformer width, W_q	0.8 mm
Length of transmission line, L_t	19.5 mm	Length of the stub, L_{stub}	16.1 mm
Feed distance, F_d	15.2 mm	Feed to patch distance, P_d	18.9 mm
Slot-line length, L_s	18 mm	Quarter wave transformer length, L_q	16.5 mm

TABLE II: Antenna Design Parameters for Creating a Dataset with Variation Range and Steps

Variable	Range (mm)	Step (mm)	Data Points
Lt	15.0 - 60.0	0.5	91
Lstub	15.0 - 20.0	0.1	51
Wt	2.0 - 3.0	0.1	11
Fd	15.0 - 20.0	0.2	26
Pd	18.0 - 22.0	0.1	41
Lm	13.0 - 18.0	0.1	51
Wq	0.3 - 1.0	0.1	8
Lq	15.0 - 18.0	0.1	31
Ws	0.1 - 0.3	0.1	3
Ls	15.0 - 20.0	0.1	51
WLsp	20.0 - 35.0	1.0	16

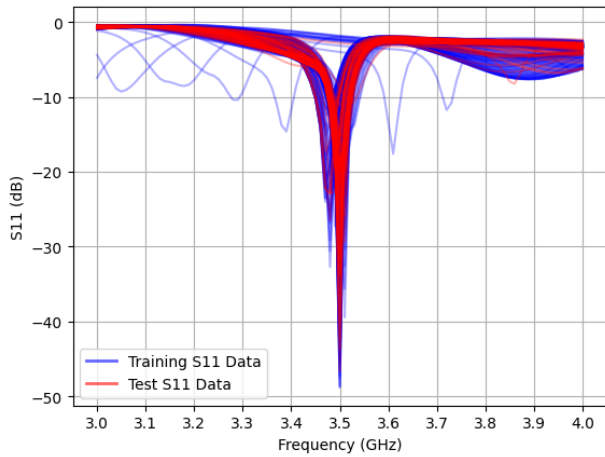


Fig. 4: All the return loss values after dataset creation.

loss values for each sample. This work aims to learn a function $f(\mathbf{X})$ that minimizes the error between the predicted return loss values $\hat{Y}$ and the actual values Y .

$$\hat{Y} = f(X) \quad (1)$$

where $\hat{Y}$ represents the predicted return loss. The task is to minimize a loss function, for instance, the Mean Squared Error (MSE), defined as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

Here, n is the number of samples in the dataset, y_i is the actual return loss, and $\hat{y}_i$ is the predicted return loss. This work aims to train different machine learning models $f_1, f_2, \dots, f_8$

to minimize this error and accurately predict the return loss based on the antenna design parameters.

For each machine learning model, a function f is aimed to be found such that:

$$f(X) \approx Y \quad (3)$$

To find this function $f(\mathbf{X})$, three linear regression models are used: 1) Linear Regression, 2) Lasso Regression, and 3) Ridge Regression. Linear Regression tries to find the best-fitted line to describe the given data. Lasso and Ridge use regularization to penalize the coefficient to reduce the chance of model overfitting. Decision Tree Regression (DT) and Random Forest Regression (RF) are tree-based models. DT creates a tree-based structure from the given data. In RF, data is further split into subsets, where each subgroup creates a separate DT to predict, and finally, this prediction is averaged to the final prediction score.

The prediction of a new sample by RF is given by:

$$\hat{Y} = \frac{1}{T} \sum_{t=1}^T h_t(X) \quad (4)$$

where T is the number of trees and $h_t(X)$ is the prediction of the t -th tree.

As shown in Equation 4, the prediction is the average of the predictions from all the trees. This work also uses one boosting algorithm, Extreme Gradient Boosting Regression (XGB), and two Bayesian Regression.

D. Hyper-parameter Tuning

Unlike the model parameter, which finds its optimum value during training, the hyperparameters are given before training. Finding the proper set of hyperparameters is essential for achieving the best result. This work uses two hyperparameter search methods: Grid search and Randomized Search. The objective function used to find the hyperparameter is:

$$\text{Objective Function} = \min_{\theta} \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i(\theta))^2 \quad (5)$$

where:

y_i is the true output,

$\hat{y}_i(\theta)$ is the predicted output based on the model parameters θ ,

n is the number of samples.

Both search methods attempt to minimize this objective function of Equation 5. During grid search, it searches across

TABLE III: Hyper-parameter Tuning Search Grid for Different Models

Model	Parameter Name	Parameter Values	Grid Search	Randomized Search	Best Tuned
Linear Regression	-	-	-	-	-
Lasso Regression	<i>alpha</i> <i>max_iter</i>	[0.0001, 0.001, 0.01, 0.1, 1, 10] [1000, 5000, 10000, 20000]	0.01 1000	0.01 1000	0.01 1000
Ridge Regression	<i>alpha</i> <i>solver</i>	[0.0001, 0.001, 0.01, 0.1, 1, 10, 100] ['auto', 'svd', 'cholesky', 'lsqr', 'sag', 'saga']	1 'lsqr'	1 'lsqr'	1 'lsqr'
Decision Tree Regression	<i>max_depth</i> <i>min_samples_split</i> <i>min_samples_leaf</i> <i>max_features</i>	[None, 5, 10, 20, 50] [2, 5, 10, 15] [1, 2, 4, 6] ['auto', 'sqrt', 'log2', None]	None 2 2 'sqrt'	20 2 2 None	20 2 2 'sqrt'
Random Forest Regression	<i>n_estimators</i> <i>max_depth</i> <i>min_samples_split</i> <i>min_samples_leaf</i> <i>bootstrap</i>	[100, 300, 500] [None, 10, 20, 30] [2, 5, 10] [1, 2, 4] [True, False]	300 20 2 1 True	300 20 2 1 True	300 20 2 1 True
XGB Regression	<i>n_estimators</i> <i>learning_rate</i> <i>max_depth</i> <i>subsample</i> <i>colsample_bytree</i>	[100, 300, 500] [0.01, 0.05, 0.1, 0.2] [3, 5, 7, 10] [0.6, 0.8, 1.0] [0.6, 0.8, 1.0]	200 0.05 7 0.6 0.6	100 0.1 7 0.6 0.6	200 0.05 7 0.6 0.6
Bayesian Linear Regression	<i>alpha_1</i> <i>alpha_2</i> <i>lambda_1</i> <i>lambda_2</i>	[1e-06, 1e-05, 1e-04, 1e-03] [1e-06, 1e-05, 1e-04, 1e-03] [1e-06, 1e-05, 1e-04, 1e-03] [1e-06, 1e-05, 1e-04, 1e-03]	- - - -	- - - -	1e-06 1e-06 1e-06 1e-06
Gaussian Process Regression	<i>alpha</i> <i>normalize_y</i>	[1e-10, 1e-5, 1e-2, 0.1] [True, False]	0.01 True	0.1 True	0.01 True

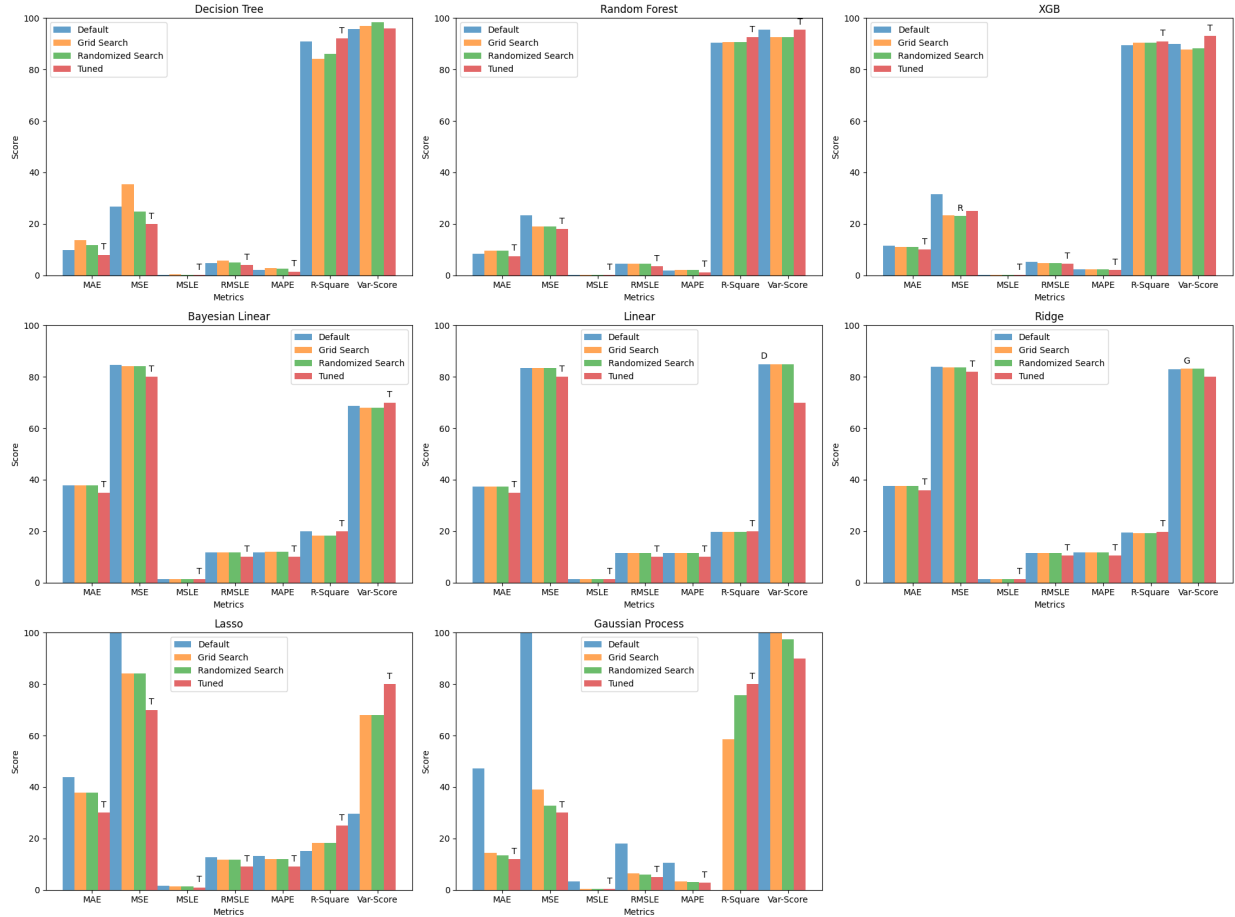


Fig. 5: Performance metrics of different algorithms.

TABLE IV: Model Performance Metrics (%)

Algorithm	MAE (%)	MSE (%)	MSLE (%)	RMSLE (%)	MAPE (%)	R-Square (%)	Var-Score (%)
Linear Regression	37.290520	83.554399	1.347530	11.608314	11.621605	19.805411	84.933223
Random Forest Regression	9.593345	18.957273	0.201523	4.489129	2.107746	90.587922	92.486653
Decision Tree Regression	14.094079	35.042813	0.320482	5.661114	3.050264	83.622927	96.102028
Lasso Regression	37.897014	84.147769	1.370619	11.707342	11.896604	18.168592	68.022816
Ridge Regression	37.497836	83.784251	1.352418	11.629350	11.667443	19.340679	83.077584
XGB Regression	11.025757	23.364249	0.237874	4.877238	2.365826	90.447176	87.697834
Gaussian Process Regression	14.478213	39.020742	0.424354	6.514243	3.240088	58.594237	119.689504

all the possible combinations of hyperparameters given in the search grid. In randomized search, the hyperparameter space is randomly sampled and then tries to find the best combination that minimizes the objective function. The total search grid for all the algorithms used and their best-found hyperparameter is given in Table III.

The best-tuned hyperparameters are found using the following objective function:

$$P_{\text{best}} = \arg \min_P E(P) \quad (6)$$

where:

- P_{best} is the best-tuned hyperparameter set,
- $E(P)$ is the evaluation metric for the model with hyperparameters P .

III. RESULT AND DISCUSSION

Several performance metrics have been used in this work to evaluate the trained model's performance: Mean Absolute Error (MAE), Mean Squared Error (MSE), Mean Squared Logarithmic Error (MSLE), Root Mean Squared Logarithmic Error (RMSLE), Mean Absolute Percentage Error (MAPE), R-Squared (R^2), and Variance Score. MAE and MSE indicate how much distance a predicted value from its actual value. Where, MSLE and RMSLE describe how well the model performed to predict relatively lower return loss values. MAPE suggests how much deviated the predicted value is. R-Square describes how much variance of the actual data is found by the model. Finally the data variability is found by Var-score.

A detailed picture of how best-tuned ML models performed and the best hyperparameter found by Grid search and randomized search is shown in Fig. 5. It is evident from the figure that all the model performed well with their tuned hyperparameter value. A detailed result for all the algorithm performance metrics is given in Table IV.

From Table IV, it is observed that **Random Forest Regression** is the best-performing model with an MAE of 9.59%, MSE of 18.96%, MAPE of 2.11%, and an R-Square of 90.59%. In XGB, a variance Score of 87.70% is slightly lower than Random Forest, but overall, its performance is better than other methods.

Decision Tree Regression performs moderately well with an MAE of 14.09% and R-Square of 83.62%. Its Variance Score of 96.10% indicates it captures most of the data's variability, though it's less accurate than Random Forest and XGB. Linear Regression with an MAE of 37.29%, MSE of 83.55%, MAPE of 11.62%, and a low R-Square of 19.81% fails to adequately

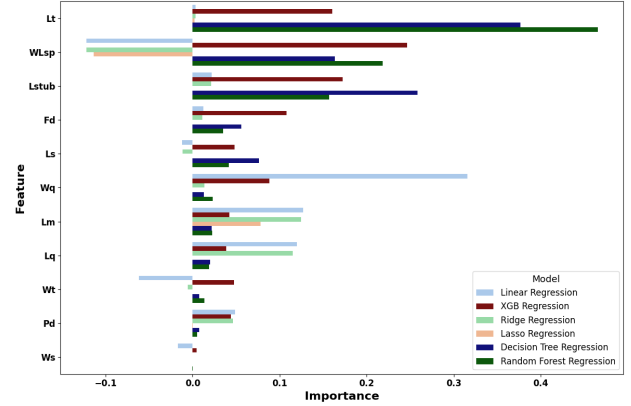


Fig. 6: Feature importance analysis.

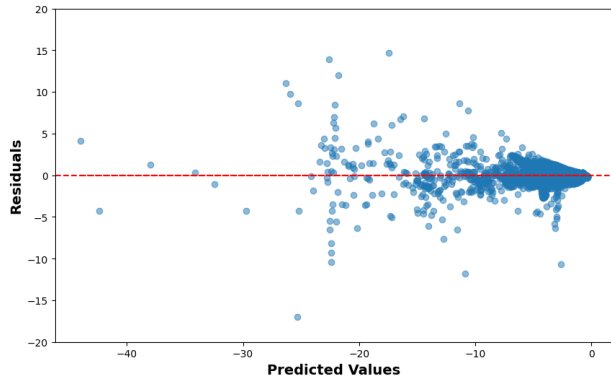
explain the variance, making it unsuitable for this task. Like linear regression, Lasso performs poorly, with an MAE of 37.90%, MSE of 84.15%, MAPE of 11.90%, and an R-Square of only 18.17%. The performance of Ridge Regression is slightly better than that of Lasso, but it is still poor.

Based on the above analysis, Random Forest Regression and XGB Regression are the most suitable models for predicting return loss. At the same time, Linear Regression, Lasso Regression, and Ridge Regression show poor predictive power.

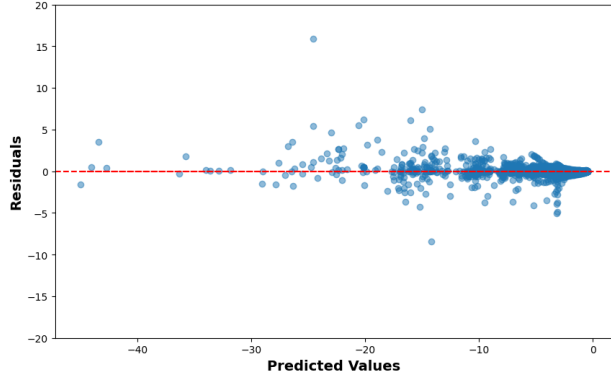
1) *Performance Analysis:* The reason why some ML models are better than others can be viewed by their feature importance analysis and residual analysis. For this specific task of predicting return loss, the antenna performance is strongly influenced by the transmission line length (L_t). By increasing or decreasing the transmission line, the input impedance of the antenna is changed, which affects antenna return loss performance. Similarly, the resonance frequency can be changed by changing the square patch length and width, which is also an essential parameter during design. Also, the stub length (L_{stsub}) affects the port isolation, and the distance between the feed (F_d) and Slot length (L_s) are some important parameters to choose from during the EM simulation stage.

From Fig. 6, it is evident that the best-performing models (Random Forest Regression, XGB Regression) they found L_t , $Wlsp$, L_{stsub} as the essential feature for their model tuning just like the EM simulation. and the least-performing model failed to identify these features as essential features.

The residual analysis can also confirm this, which is a difference between the predicted and actual values. From the least performing model Lasso regression, the residual



(a) Residual of Lasso Regression



(b) Residual of Random Forest Regression

Fig. 7: Residual analysis of Regression Models.

plot in Fig. 7a the residuals fan out as predicted values increase, indicating heteroscedasticity (non-constant variance) and inability to find the data's complexity. whereas for best best-performing model Random forest regression in Fig. 7b, the residuals are randomly scattered around the zero line, indicating homoscedasticity (constant variance) and a better fit to the data.

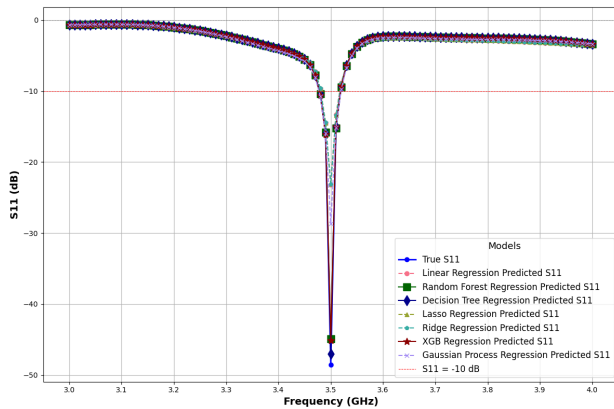


Fig. 8: Performance metrics of different algorithms.

The overall performance of the tuned ML models for the best-performing design in EM simulation is shown in Fig. 8.

IV. CONCLUSION

This work comprehensively analyzes how machine learning models can be used and tuned to predict antenna performance. For this, a 3.5GHz antenna was designed, and a dataset of 380 samples was created. Eight regression Models and two search optimization methods were used for hyperparameter optimization. Seven error metrics were used to evaluate the model performance. The model performance was validated using feature importance and residual analysis. Out of eight machine learning models, the Random Forest Regression outperforms all of them by an MAE of 9.59%, MSE of 18.96%, MAPE of 2.11%, and an R-Square of 90.59% and Variance Score of 92.49%. For future studies, these ML models can be used to optimize the design parameters to enhance the antenna's performance further.

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